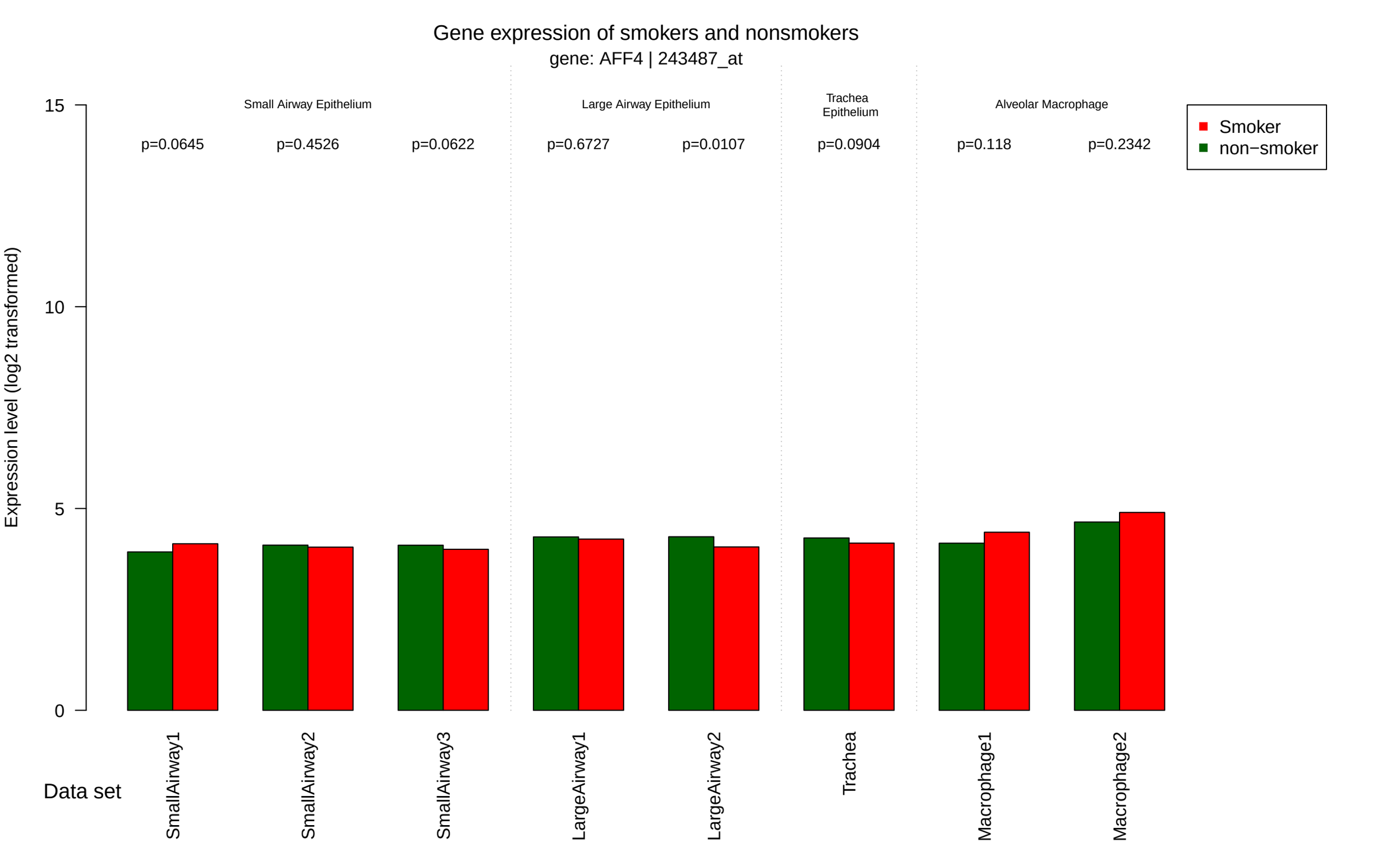
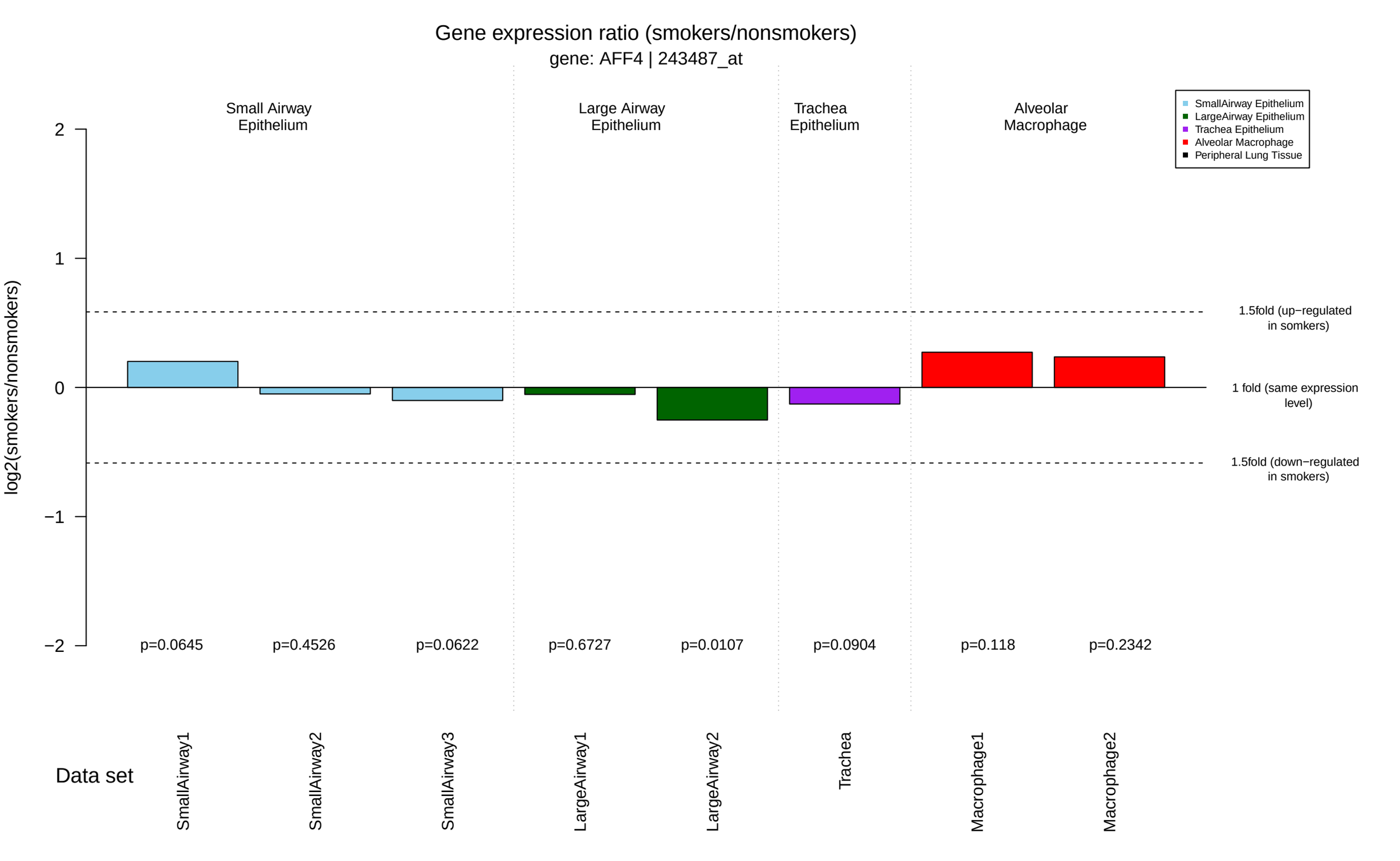
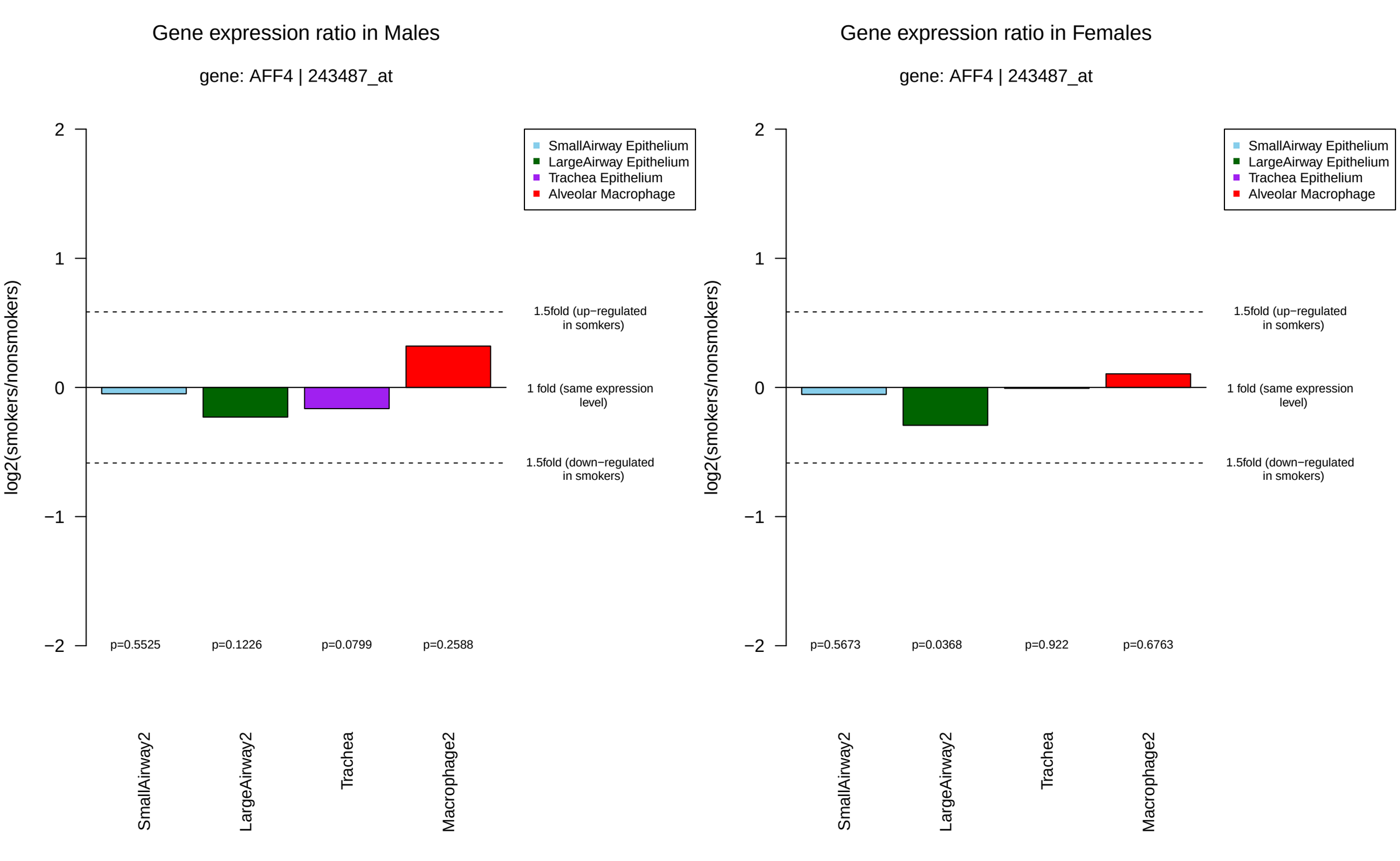
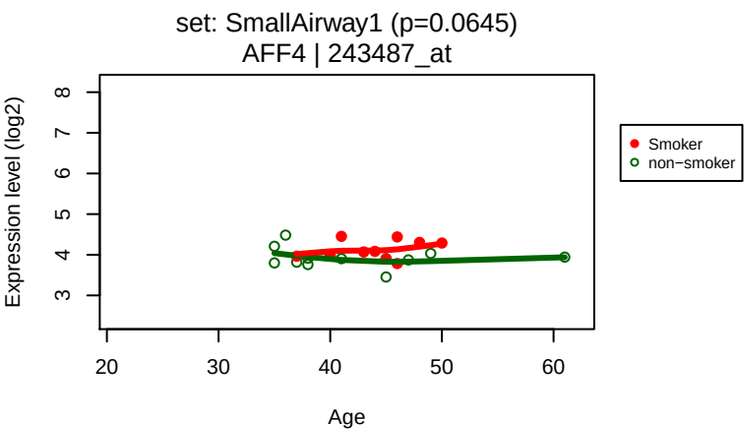
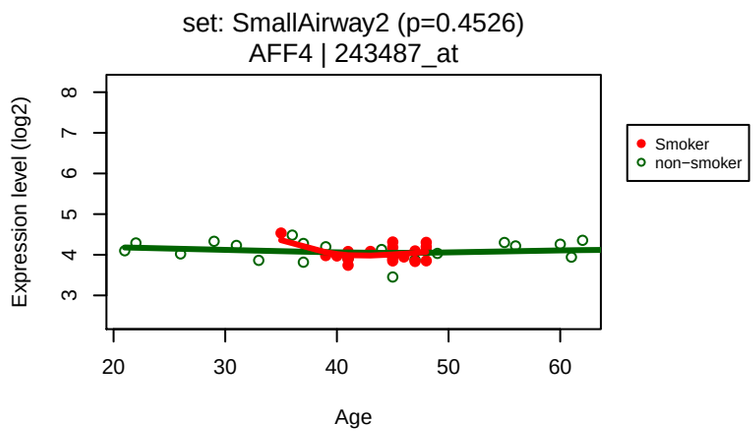
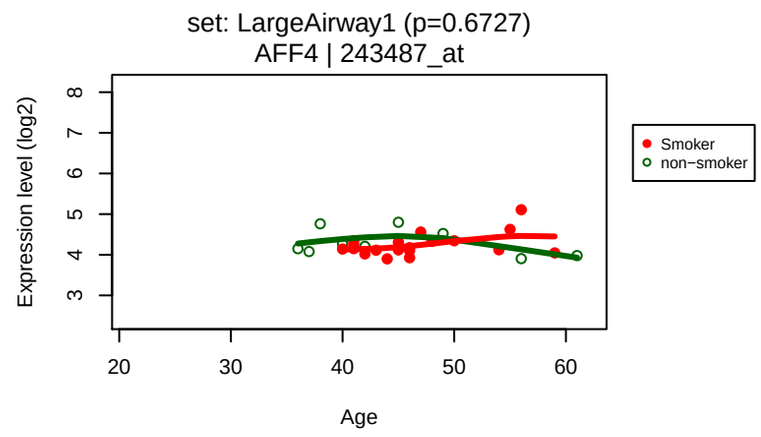

Gene expression of smokers and nonsmokers
gene: AFF4 | 243487_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0645
p=0.4526
p=0.0622
p=0.6727
p=0.0107
p=0.0904
p=0.118
p=0.2342
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

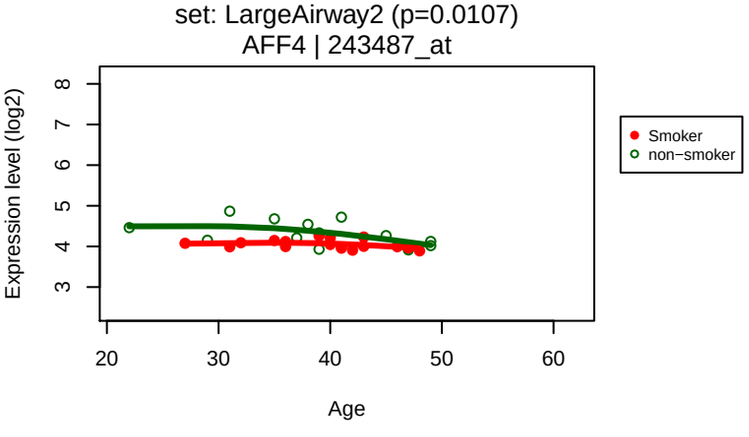

Gene expression ratio (smokers/nonsmokers)
gene: AFF4 | 243487_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0645
p=0.4526
p=0.0622
p=0.6727
p=0.0107
p=0.0904
p=0.118
p=0.2342
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

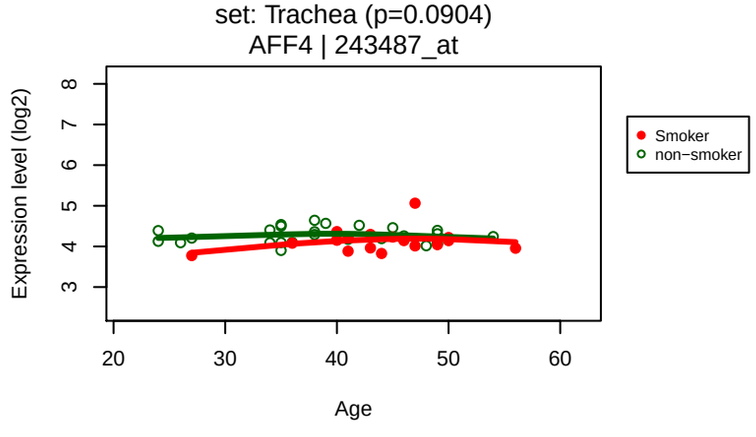

Gene expression ratio in Males
gene: AFF4 | 243487_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5525
p=0.1226
p=0.0799
p=0.2588
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AFF4 | 243487_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5673
p=0.0368
p=0.922
p=0.6763
SmallAirway2
LargeAirway2
Trachea
Macrophage2

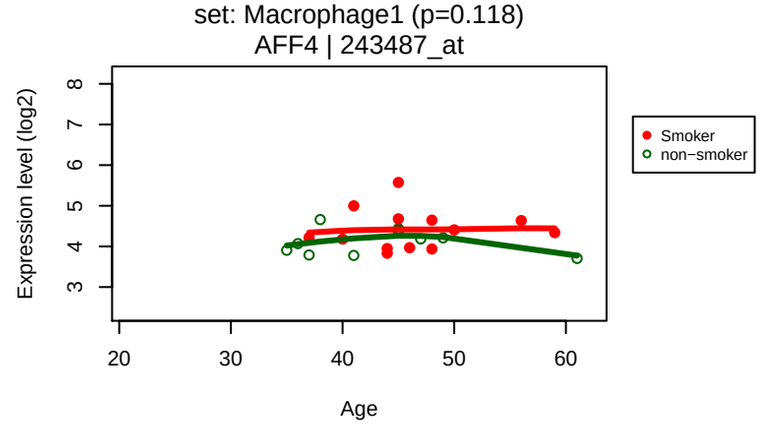

set: SmallAirway1 (p=0.0645)
AFF4 | 243487_at
Expression level (log2)
Age
Smoker
non−smoker

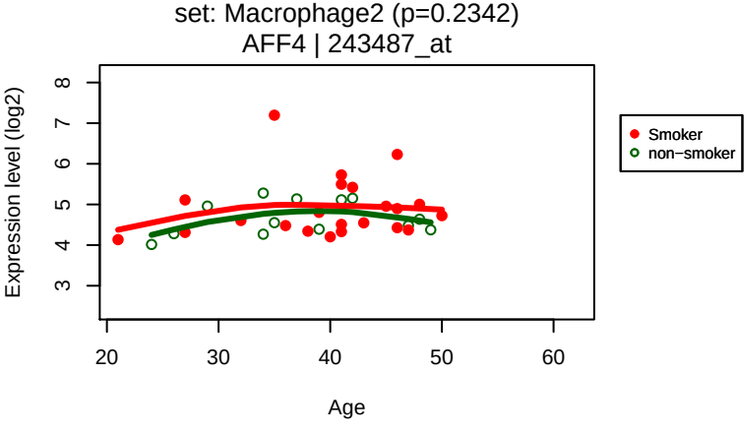

set: SmallAirway2 (p=0.4526)
AFF4 | 243487_at
Expression level (log2)
Age
Smoker
non−smoker

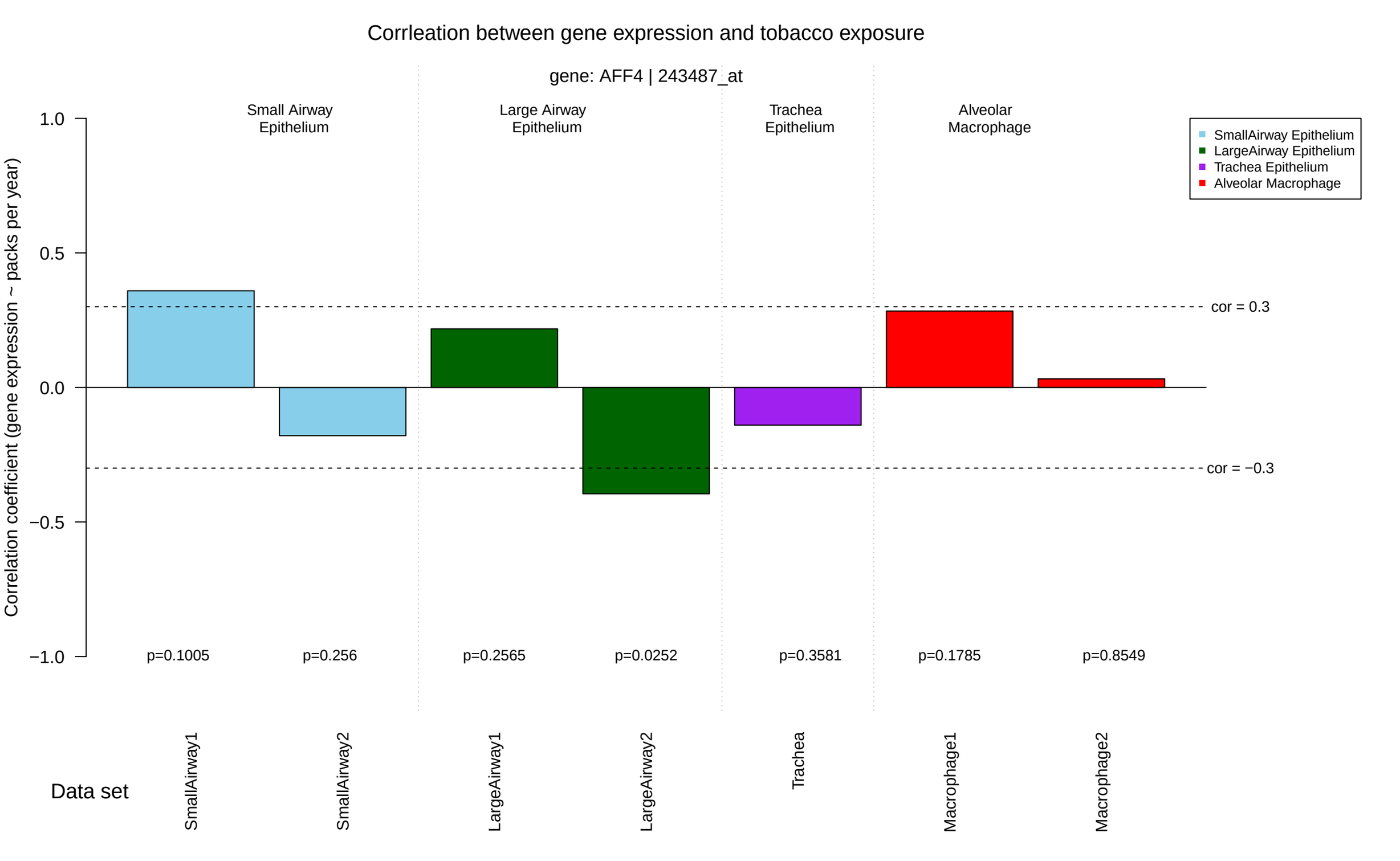

set: LargeAirway1 (p=0.6727)
AFF4 | 243487_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0107)
AFF4 | 243487_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0904)
AFF4 | 243487_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.118)
AFF4 | 243487_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2342)
AFF4 | 243487_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AFF4 | 243487_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1005
p=0.256
p=0.2565
p=0.0252
p=0.3581
p=0.1785
p=0.8549
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2